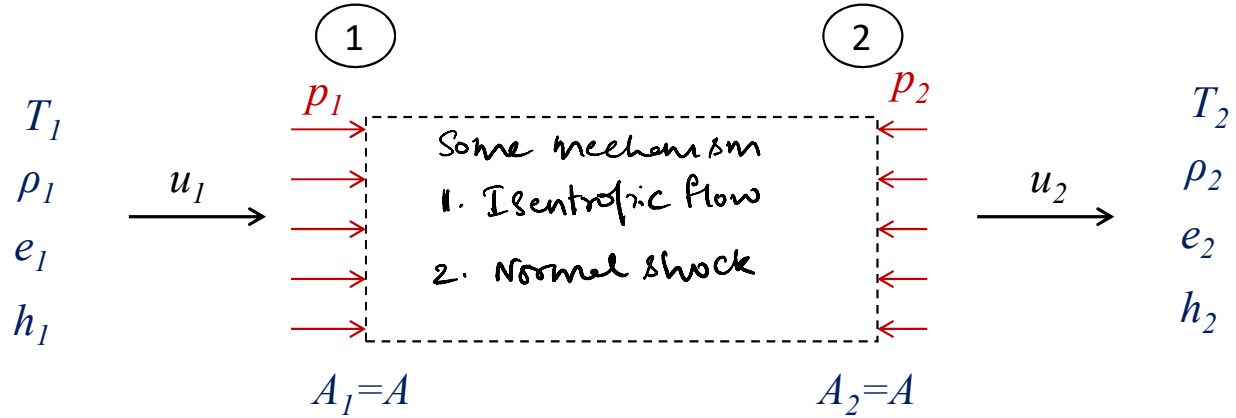
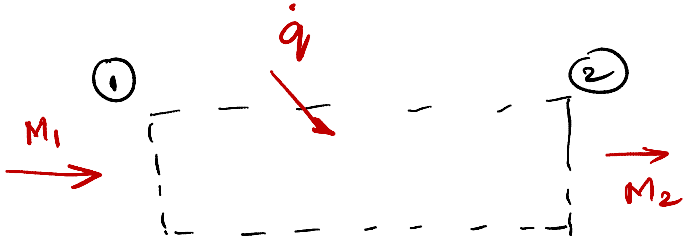
One-dimensional flows



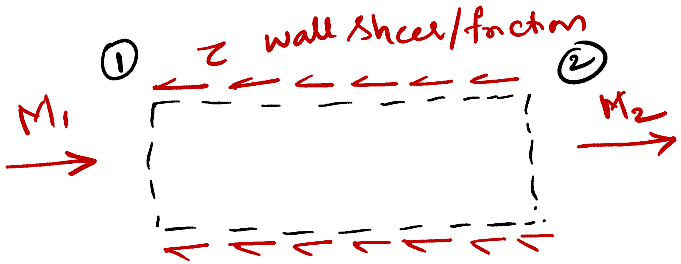
Unit-4

1. 1-D flow with heat transfer (heat addition or removal)
   - RAYLEIGH FLOW



Eg: Combustion chamber of a gas turbine engine

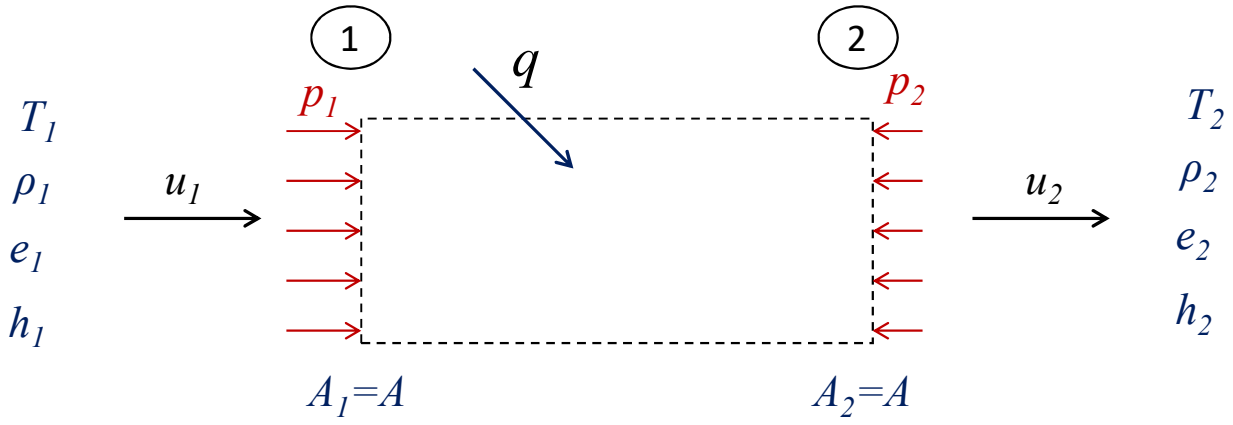
2. 1-D flow with friction (or) wall sheer
   - FANNO FLOW



Eg: Flow of gases through long pipes

ONE-DIMENSIONAL FLOW WITH HEAT ADDITION

Rayleigh Flow



Continuity: $$\rho_1 u_1 = \rho_2 u_2$$

Momentum: $$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

Energy: $$q + h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

We can further simplify these equations for a calorically perfect gas.

$$q = \left(c_p T_2 + \frac{u_2^2}{2}\right) - \left(c_p T_1 + \frac{u_1^2}{2}\right) = c_p\left(T_2 + \frac{u_2^2}{2c_p}\right) - c_p\left(T_1 + \frac{u_1^2}{2c_p}\right)$$

T_{02} $\qquad$ T_{01}

$$\boxed{q = c_p T_{o_2} - c_p T_{o_1} = c_p\left(T_{o_2} - T_{o_1}\right)}$$

- Earlier, when the flow is adiabatic $T_{01} = T_{02}$
- For Non-adiabatic flow $q = c_p(T_{02} - T_{01})$

***The effect of heat addition is to directly change the total temperature of the flow.

***If heat is added, T_0 increases. If heat is extracted, T_0 decreases.

Let us proceed to find the ratios of properties between regions 1 and 2 in terms of the Mach numbers M_1 and M_2 .

$\frac{p_2}{p_1}$: Momentum eqn: $p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$

$$\Rightarrow p_2 - p_1 = \rho_1 u_1^2 - \rho_2 u_2^2$$

$\rho u^2 = \rho a^2 M^2 = \rho \frac{\gamma p}{\rho} M^2$ $\quad (\because u = Ma,\ a = \sqrt{\frac{\gamma p}{\rho}})$

$= \gamma p M^2$

$$p_2 - p_1 = \gamma p_1 M_1^2 - \gamma p_2 M_2^2$$

$\Rightarrow p_2 (1 + \gamma M_2^2) = p_1 (1 + \gamma M_1^2)$

$\Rightarrow$

$$\boxed{\frac{p_2}{p_1} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}} \quad \text{——} (1)$$

$\frac{T_2}{T_1}$: Ideal gas law $T = p/\rho R$

From Continuity $\rho_1 u_1 = \rho_2 u_2$

$$\frac{T_2}{T_1} = \frac{p_2}{p_1}\frac{\rho_1}{\rho_2} = \frac{p_2}{p_1}\frac{u_2}{u_1}$$

$$\frac{u_2}{u_1} = \frac{M_2}{M_1}\frac{a_2}{a_1} = \frac{M_2}{M_1}\left(\frac{T_2}{T_1}\right)^{1/2} \quad (\because a = \sqrt{\gamma R T})$$

$\Rightarrow \frac{a_2}{a_1} = \sqrt{\frac{T_2}{T_1}}$

$$\boxed{\frac{T_2}{T_1} = \left(\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}\right)^2 \left(\frac{M_2}{M_1}\right)^2} \quad \text{——} (2)$$

Since $\rho_2/\rho_1 = (p_2/p_1)(T_1/T_2)$,

$$\boxed{\frac{\rho_2}{\rho_1} = \left(\frac{1 + \gamma M_2^2}{1 + \gamma M_1^2}\right)\left(\frac{M_1}{M_2}\right)^2} \quad \text{——} (3)$$

Stagnation conditions:

$\frac{p_{02}}{p_{01}}$:

$$\frac{P_{02}}{P_{01}} = \frac{P_{02} \times \frac{P_2}{P_2}}{P_{01} \times \frac{P_1}{P_1}} = \frac{\frac{P_{02}}{P_2}}{\frac{P_{01}}{P_1}} \times \frac{P_2}{P_1}$$

From isentropic relation $\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}$

$$\frac{P_{02}}{P_{01}} = \left[\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2}\right]^{\frac{\gamma}{\gamma-1}} \times \left[\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}\right]$$

$$\boxed{\frac{p_{o_2}}{p_{o_1}} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2}\right)^{\gamma/(\gamma-1)}} \quad (4)$$

$\frac{T_{02}}{T_{01}}$:

$$\frac{T_{02}}{T_{01}} = \frac{\left(\frac{T_{02}}{T_2}\right)}{\left(\frac{T_{01}}{T_1}\right)} X \frac{T_2}{T_1}$$

From isentropic relations $\frac{T_0}{T} = \left(1 + \frac{\gamma-1}{2} M^2\right)$

$$\frac{T_{02}}{T_{01}} = \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2}\right) \left[\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}\right]^2 \left(\frac{M_2}{M_1}\right)^2$$

$$\boxed{\frac{T_{o_2}}{T_{o_1}} = \left(\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}\right)^2 \left(\frac{M_2}{M_1}\right)^2 \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2}\right)} \quad (5)$$

** $\frac{p_2}{p_1}, \frac{T_2}{T_1}, \frac{\rho_2}{\rho_1}, \frac{p_{02}}{p_{01}}, \frac{T_{02}}{T_{01}} = f(M_1, M_2)$

solution procedure for one-dimensional flow with heat addition:

- All conditions in region 1 are given. Amount of heat added or removed q is given. We have to find conditions in region 2.

Knowns: $M_1, p_1, T_1, p_{o1}, T_{o1}, q$

Find: $M_2, p_2, T_2, p_{o2}, T_{o2}$

- Therefore, for a given *q*,

$$q = c_p(T_{02} - T_{01})$$

$$\Rightarrow T_{02} = \frac{q}{c_p} + T_{01}$$

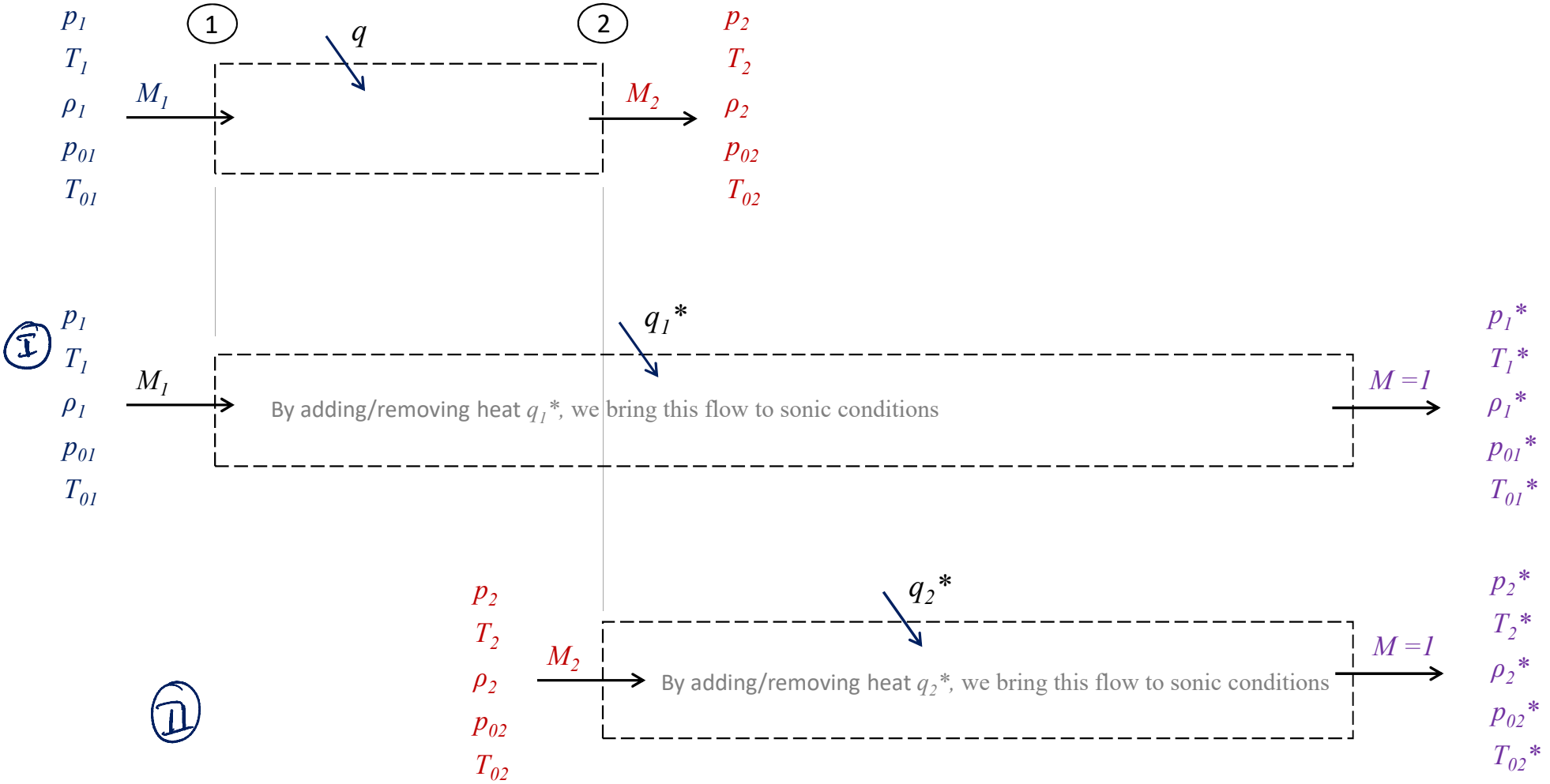
- Use $\frac{T_{o2}}{T_{o1}} = \left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right)^2 \left(\frac{M_2}{M_1}\right)^2 \left(\frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}\right)$ to find M_2

Problem in this procedure:
NO direct solution for M_2. We have to use trail-and-error method to obtain M_2.

- After finding M_2, find p_2/p_1, T_2/T_1, ρ_2/ρ_1, p_{02}/p_{01}, etc from the derived formulae

$$\frac{p_2}{p_1}, \frac{T_2}{T_1}, \frac{p_{o2}}{p_{o1}} = f(M_1, M_2)$$

We don't use this method because of the above problem


1
2
q
p_1
T_1
M_1
ρ_1
p_{01}
T_{01}
p_2
T_2
M_2
ρ_2
p_{02}
T_{02}
I
q_1*
p_1
T_1
M_1
ρ_1
p_{01}
T_{01}
By adding/removing heat q_1*, we bring this flow to sonic conditions
$M = 1$
p_1*
T_1*
ρ_1*
p_{01}*
T_{01}*
II
q_2*
p_2
T_2
M_2
ρ_2
p_{02}
T_{02}
By adding/removing heat q_2*, we bring this flow to sonic conditions
$M = 1$
p_2*
T_2*
ρ_2*
p_{02}*
T_{02}*

$$\Rightarrow \frac{p_2}{p_1} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2} \quad \text{— (1)}$$

(I) $M_1 = M_1$, $p_1 = p_1$
$M_2 = 1$, $p_2 = p_1^*$

$$\frac{p_1^*}{p_1} = \frac{1+\gamma M_1^2}{1+\gamma(1)} \quad \text{— (A)}$$

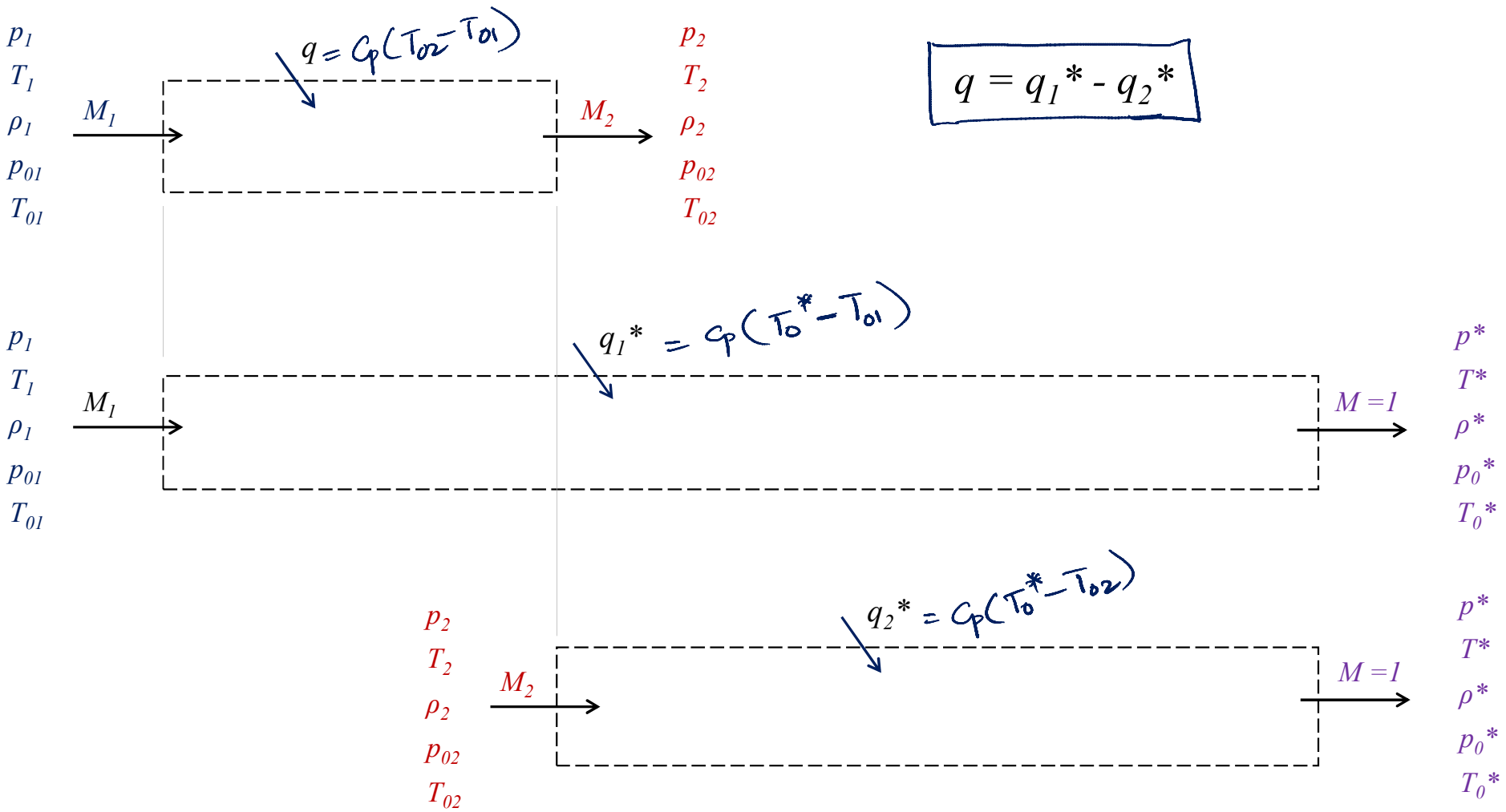
(II) $M_1 = M_2$, $p_1 = p_2$
$M_2 = 1$, $p_2 = p_2^*$

$$\frac{p_2^*}{p_2} = \frac{1+\gamma M_2^2}{1+\gamma(1)} \quad \text{— (B)}$$

$$\frac{(A)}{(B)} \Rightarrow \frac{p_2}{p_1} \times \frac{p_1^*}{p_2^*} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2} \quad \text{— (C)}$$

Comparing (1) & (C) $\boxed{p_1^* = p_2^*}$

- Sonic conditions remains same in the 1-D flow with heat transfer


p_1
T_1
ρ_1
p_{01}
T_{01}
M_1
$q = c_p(T_{02} - T_{01})$
M_2
p_2
T_2
ρ_2
p_{02}
T_{02}
$q = q_1^* - q_2^*$
p_1
T_1
ρ_1
p_{01}
T_{01}
M_1
$q_1^* = c_p(T_0^* - T_{01})$
$M = 1$
p^*
T^*
ρ^*
p_0^*
T_0^*
p_2
T_2
ρ_2
p_{02}
T_{02}
M_2
$q_2^* = c_p(T_0^* - T_{02})$
$M = 1$
p^*
T^*
ρ^*
p_0^*
T_0^*

- For convenience of calculation, we use sonic flow as a reference condition
- Note that, for a given flow, no matter what the local flow properties are, the reference sonic conditions (the starred quantities) are constant values
- Here, p*, T* , p_0* and T_0* are those conditions in a one-dimensional flow that would exist if enough heat is added to achieve Mach 1

In eqns ① - ⑤

$p_1 = p$, $M_1 = M$

$p_2 = 1$, $M_2 = 1$

Substitute

$$\frac{p}{p^*} = \frac{1+\gamma}{1+\gamma M^2}$$

$$\frac{T}{T^*} = M^2\left(\frac{1+\gamma}{1+\gamma M^2}\right)^2$$

$$\frac{\rho}{\rho^*} = \frac{1}{M^2}\left(\frac{1+\gamma M^2}{1+\gamma}\right)$$

$$\frac{p_o}{p_o^*} = \frac{1+\gamma}{1+\gamma M^2}\left[\frac{2+(\gamma-1)M^2}{\gamma+1}\right]^{\gamma/(\gamma-1)}$$

$$\frac{T_o}{T_o^*} = \frac{(\gamma+1)M^2}{(1+\gamma M^2)^2}[2+(\gamma-1)M^2]$$



* Sonic reference conditions

p^*, T^*, ρ^*, p_0^*, T_0^*

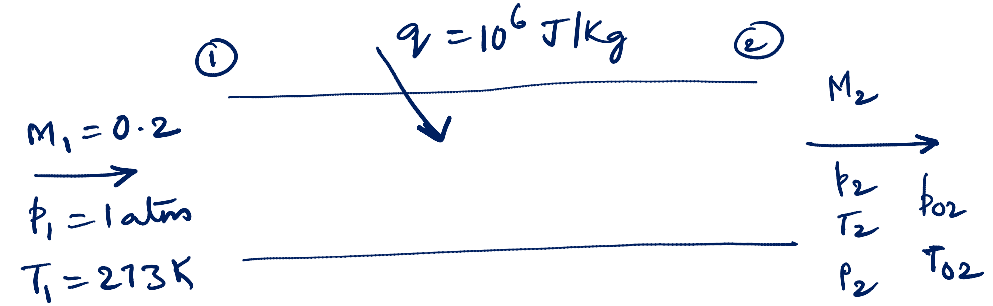
These are constant in the flow.

TABLE 8.2

Flow of perfect gases with heat transfer ($\gamma = 1.4$)
(Rayleigh flow)

M	p/p^*	p_0/p_0^*	T/T^*	T_0/T_0^*	$c/c^* = \rho^*/\rho$
0.62	1.560	1.068	0.935	0.839	0.599
0.64	1.525	1.061	0.953	0.859	0.625
0.66	1.490	1.055	0.968	0.877	0.650
0.68	1.457	1.049	0.981	0.893	0.673
0.70	1.423	1.043	0.993	0.908	0.697

Air enters a constant-area duct at $M_1 = 0.2$, $p_1 = 1$ atm, and $T_1 = 273$ K. Inside the duct, the heat added per unit mass is $q = 1.0 \times 10^6$ J/kg. Calculate the flow properties M_2, p_2, T_2, ρ_2, T_{o_2}, and p_{o_2} at the exit of the duct.



Sol:

Step 1: find p_{o_1}, T_{o_1}

Flow is locally isentropic at ①

For $M_1 = 0.2$ $\xrightarrow[\text{tables}]{\text{Isentropic flow}}$ $\frac{p_1}{p_{o_1}} = 0.973$ $\frac{T_1}{T_{o_1}} = 0.992$

$\Rightarrow p_{o_1} = 1.028$ atm $\quad T_{o_1} = 275.2$ K

Step 2: $q = c_p (T_{o_2} - T_{o_1})$

$$\Rightarrow T_{o_2} = \frac{q}{c_p} + T_{o_1} = \frac{10^6}{1005} + 275.2 = 1270 \text{ K}$$

Step 3: Find Sonic Conditions at ①

For $M_1 = 0.2$ $\xrightarrow[\text{tables}]{\text{Rayleigh flow}}$ $\frac{p_1}{p^*} = 2.273$ $\frac{T_1}{T^*} = 0.207$ $\frac{p_{01}}{p_0^*} = 1.235$ $\frac{T_{01}}{T_0^*} = 0.174$

$\Rightarrow p^* = 0.44 \text{ atm}$ $T^* = 1318.8 \text{ K}$ $p_0^* = 0.832 \text{ atm}$ $T_0^* = 1581.6 \text{ K}$

Sonic Conditions are Constant in the entire flow

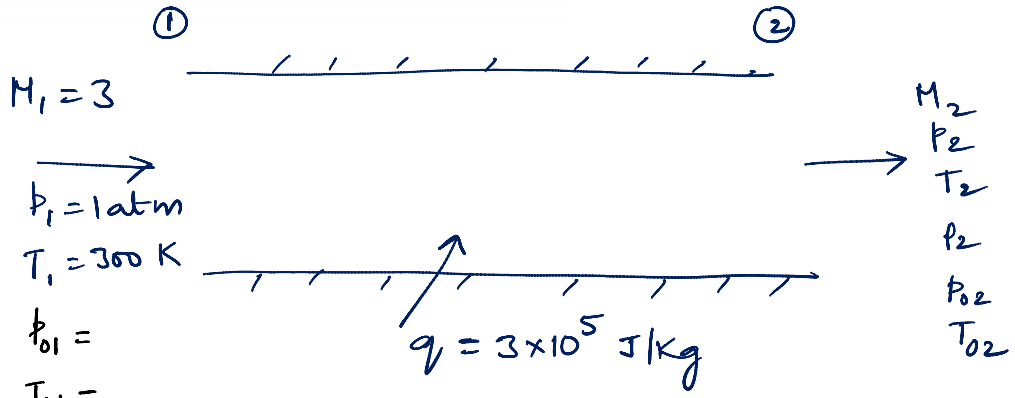
Step 4: $\frac{T_{02}}{T_0^*} = \frac{1270}{1581.6} = 0.803$

Step 5: For $\frac{T_{02}}{T_0^*} = 0.803$ $\xrightarrow{\text{Rayleigh flow tables}}$ $M_2 = 0.58$ $\frac{p_2}{p^*} = 1.632$ $\frac{T_2}{T^*} = 0.896$ $\frac{p_{02}}{p_0^*} = 1.082$

$\Rightarrow p_2 = 0.718 \text{ atm}$ $T_2 = 1183 \text{ K}$ $p_{02} = 0.902 \text{ atm}$

$\rho_2 = \frac{p_2}{RT_2} = \frac{0.718 \times 101325}{287 \times 1183} =$ kg/m^3

Air enters a constant-area duct at $M_1 = 3$, $p_1 = 1$ atm, and $T_1 = 300$ K. Inside the duct, the heat added per unit mass is $q = 3 \times 10^5$ J/kg. Calculate the flow properties M_2, p_2, T_2, ρ_2, T_{o_2}, and p_{o_2} at the exit of the duct.



At ①, flow is locally isentropic

1: For $M_1 = 3$ $\xrightarrow[\text{flow tables}]{\text{Isentropic}}$ $\frac{p_1}{p_{01}} = 0.0272$, $\frac{T_1}{T_{01}} = 0.357$

$\Rightarrow p_{01} = 36.76$ atm

$T_{01} = 840$ K

we know $q = C_p(T_{02} - T_{01})$

$$\Rightarrow T_{02} = \frac{q}{C_p} + T_{01} = \frac{3 \times 10^5}{1005} + 840 = 1139 \text{ K}$$

$$\Rightarrow T_{02} = 1139 \text{ K}$$

For $M_1 = 3$ $\xrightarrow{\text{Rayleigh flow tables}}$ $\frac{p_1}{p^*} = 0.176$ $\quad \frac{T_1}{T^*} = 0.281$ $\quad \frac{p_{01}}{p_0^*} = 3.424$ $\quad \frac{T_{01}}{T_0^*} = 0.654$

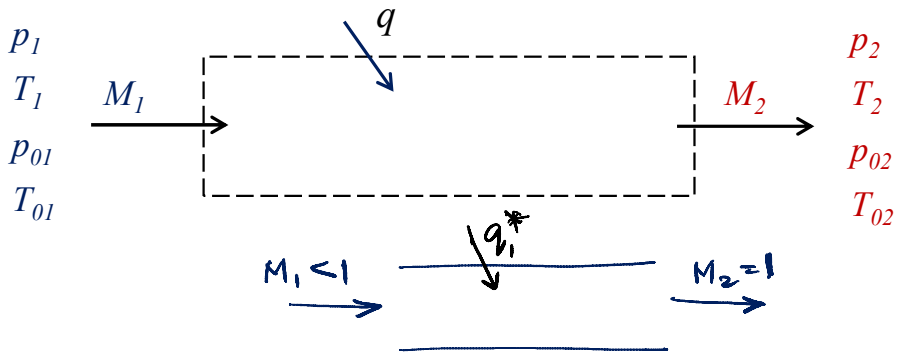
$\Rightarrow p^* = 5.682 \text{ atm}, T^* = 1067 \text{ K}$, $p_0^* = 10.75 \text{ atm}$, $T_0^* = 1284 \text{ K}$

Find $\frac{T_{02}}{T_0^*} = \frac{1139}{1284} = 0.89$

For $\frac{T_{02}}{T_0^*} = 0.89$ $\xrightarrow{\text{Rayleigh flow tables}}$ $M_2 = 1.58$ $\quad \frac{p_2}{p^*} = 0.534$ $\quad \frac{T_2}{T^*} = 0.711$ $\quad \frac{p_{02}}{p_0^*} = 1.164$

$\Rightarrow p_2 = 3.025 \text{ atm}$ $\quad T_2 = 759 \text{ K}$ $\quad p_{02} = 12.5 \text{ atm}$

$$\rho_2 = \frac{p_2}{RT_2} = \frac{3.025 \times 101325}{287 \times 759} = 1.41 \quad \text{kg/m}^3$$

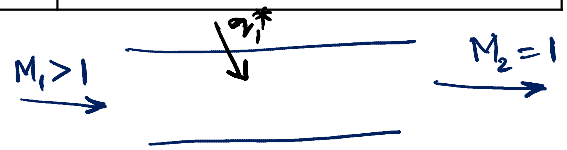


(+ve)

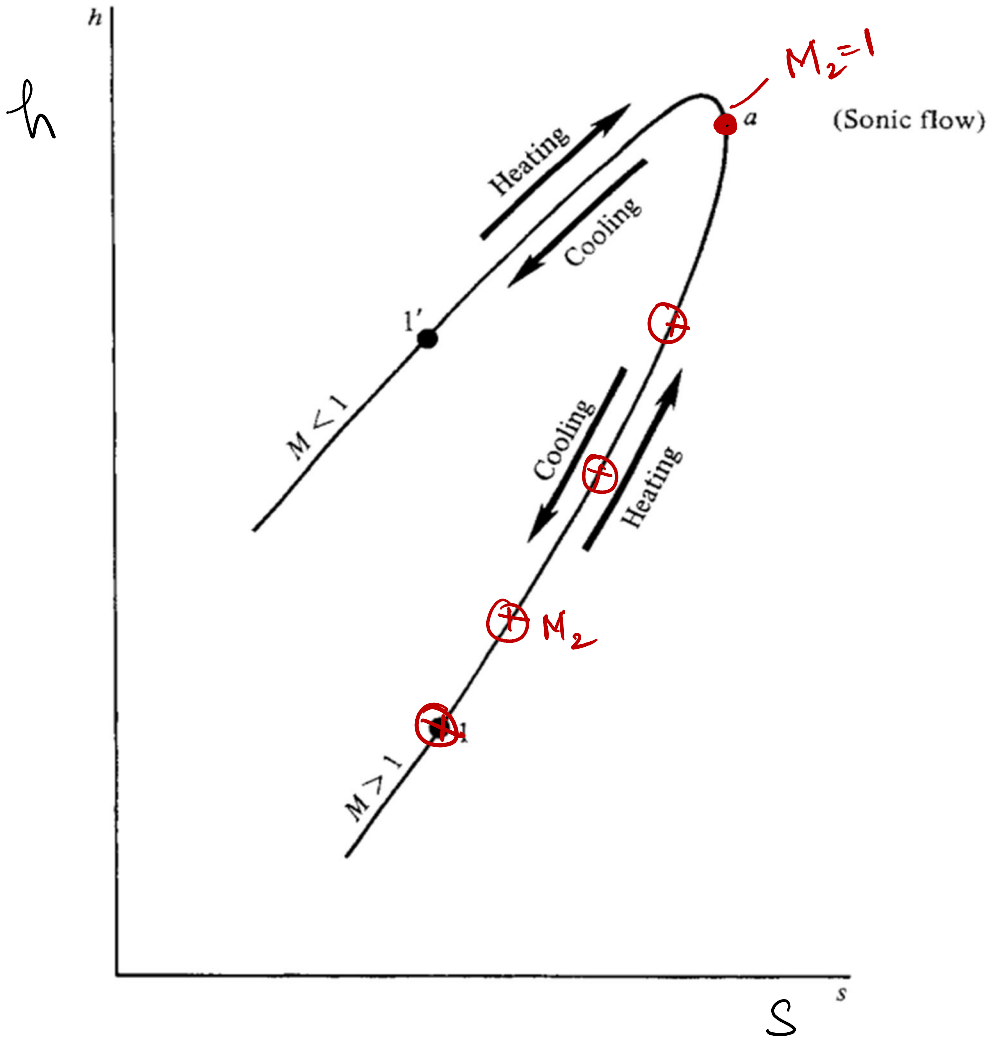
$q = C_p (T_{02} - T_{01})$

Parameter change between 1 and 2 because of heat addition ✓

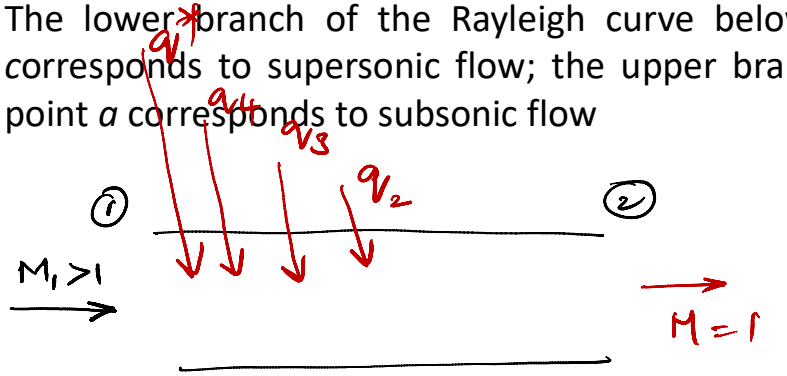
	Flow at station (1)	
	M_1 is subsonic	M_1 is supersonic
Mach Number	Increases $M_2 > M_1$	Decreases $M_2 < M_1$
Pressure	Decreases $p_2 < p_1$	Increases $p_2 > p_1$
Temperature	Increases $T_2 > T_1$	Increases $T_2 > T_1$
Stagnation pressure	Decreases $p_{02} < p_{01}$	Decreases $p_{02} < p_{01}$
Stagnation Temperature	Increases $T_{02} > T_{01}$	Increases $T_{02} > T_{01}$



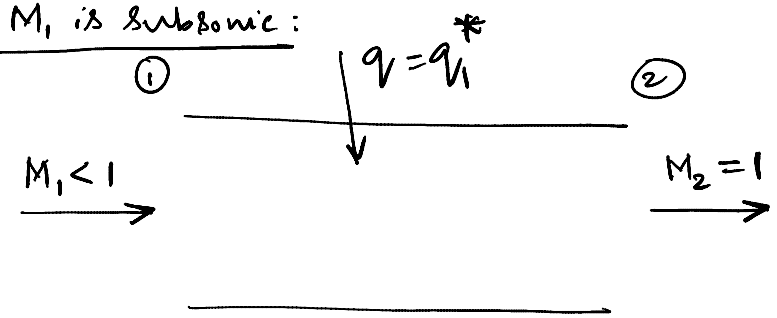
The Rayleigh curve



- Mollier diagram of 1D heat-transfer process
- If the conditions in region 1 are given by point 1, then Rayleigh curve is the locus of all possible states in region 2. Each point on the curve corresponds to a different value of *q* added or taken away.
- At Point *a* the flow becomes sonic – CHOKED CONDITION. Also, Point *a* corresponds to maximum entropy.
- The lower branch of the Rayleigh curve below point *a* corresponds to supersonic flow; the upper branch above point *a* corresponds to subsonic flow

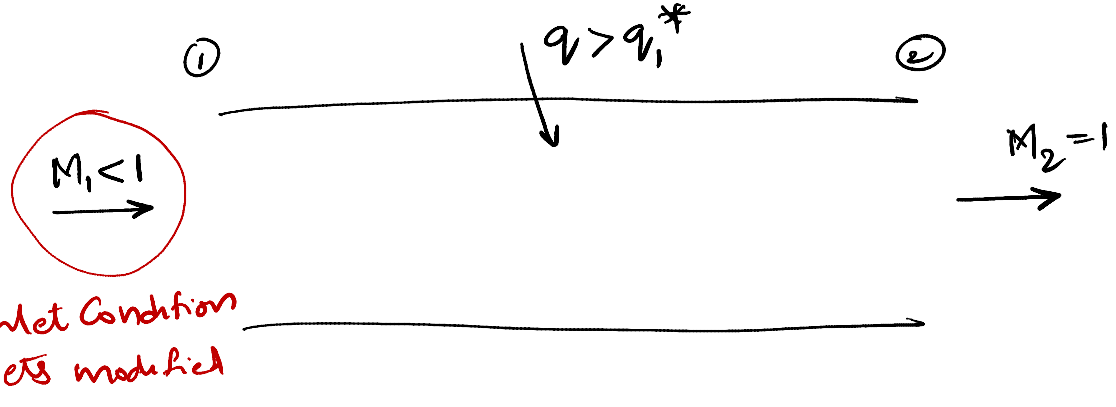


When M_1 is subsonic:



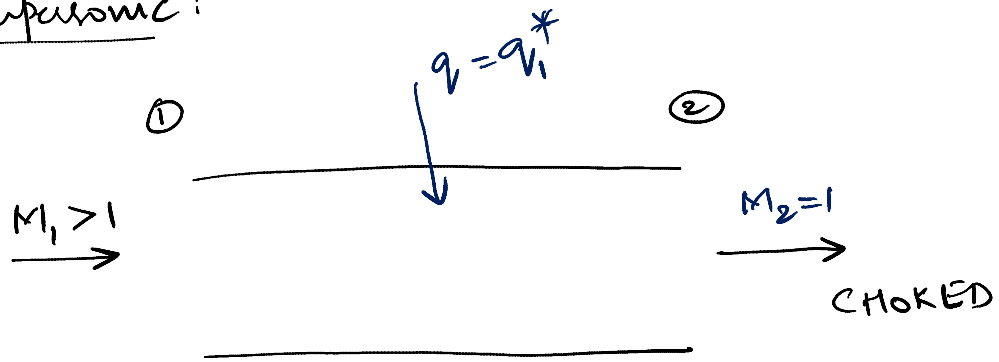
q_1^* – Heat added corresponding to the choked condition

What happens if heat added is more than q_1^* ?



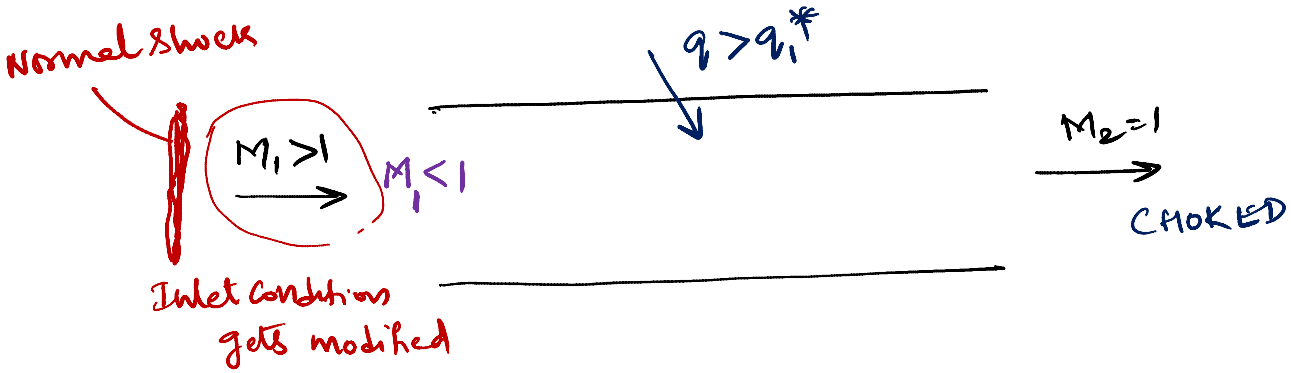
* M_1 starts decreasing when $q > q_1^*$

When M_1 is supersonic:



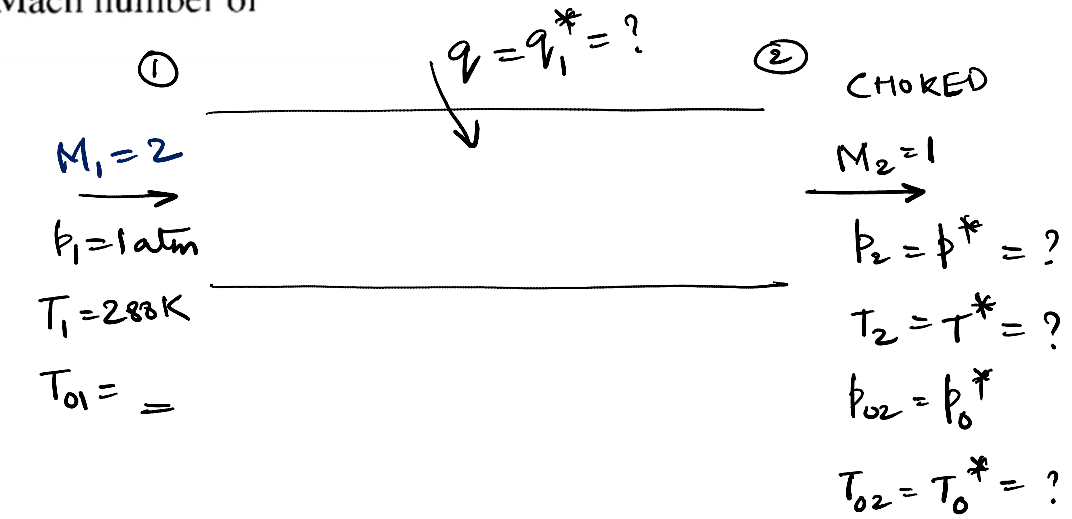
as heat added, $M_2 \downarrow$

What happens when heat added $q > q_1^*$



* Normal shock forms upstream of the inlet and the flow at inlet becomes subsonic

Consider air entering a heated duct at $p_1 = 1$ atm and $T_1 = 288$ K. Ignore the effect of friction. Calculate the amount of heat per unit mass (in joules per kilogram) necessary to choke the flow at the exit of the duct, as well as the pressure and temperature at the duct exit, for an inlet Mach number of (*a*) $M_1 = 2.0$ (*b*) $M_1 = 0.2$.



Sol:

(a) $M_1 = 2$

$$q = c_p(T_{02} - T_{01})$$

Choked flow $q_1^* = c_p(T_0^* - T_{01})$

- At ①, flow is locally isentropic

For $M_1 = 2$ $\xrightarrow[\text{tables}]{\text{Isentropic flow}}$ $\dfrac{p_1}{p_{01}} = 0.128$ $\quad \dfrac{T_1}{T_{01}} = 0.555$

$$\Rightarrow T_{01} = \frac{T_1}{0.555} = 518.9 \text{ K}$$

- For $M_1 = 2$ $\xrightarrow{\text{Rayleigh flow tables}}$ $\frac{p_1}{p^*} = 0.364$ $\quad \frac{T_1}{T^*} = 0.529$ $\quad \frac{T_{01}}{T_0^*} = 0.793$

$\Rightarrow p^* = 2.45 \text{ atm}$ $\quad T^* = 544.4 \text{ K}$ $\quad T_0^* = 654.4 \text{ K}$

> Sonic conditions p^*, T^*, p_0^*, T_0^* are constant in the entire Rayleigh flow

- Amount of heat added to choke the flow

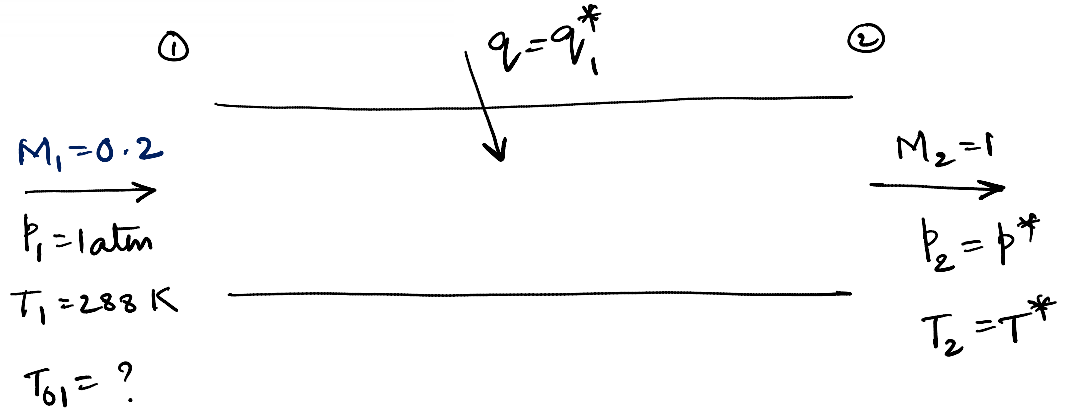
$$q_1^* = c_p (T_0^* - T_{01})$$
$$= 1005 (654.4 - 518.9)$$
$$= 136.16 \text{ KJ/kg}$$

- pressure and temperature at duct exit

$$p_2 = p^* = 2.45 \text{ atm}$$
$$T_2 = T^* = 544.4 \text{ K}$$

Consider air entering a heated duct at $p_1 = 1$ atm and $T_1 = 288$ K. Ignore the effect of friction. Calculate the amount of heat per unit mass (in joules per kilogram) necessary to choke the flow at the exit of the duct, as well as the pressure and temperature at the duct exit, for an inlet Mach number of
~~(a) $M_1 = 2.0$~~ (b) $M_1 = 0.2$.

Sol:



$$q_1^* = C_p (T_0^* - T_{01})$$

For $M_1 = 0.2$ $\xrightarrow{\text{Isentropic flow tables}}$ $\dfrac{T_1}{T_{01}} = 0.992 \Rightarrow T_{01} = \dfrac{T_1}{0.992} = 290.3$ K

For $M_1 = 0.2$ $\xrightarrow{\text{Rayleigh flow tables}}$ $\dfrac{p_1}{p^*} = 2.273$, $\dfrac{T_1}{T^*} = 0.207$, $\dfrac{T_{01}}{T_0^*} = 0.174$

$\Rightarrow p^* = 0.44$ atm $\quad T^* = 1391.3$K $\quad T_0^* = 1668.5$ K

$p_2 = p^* = 0.44$ atm

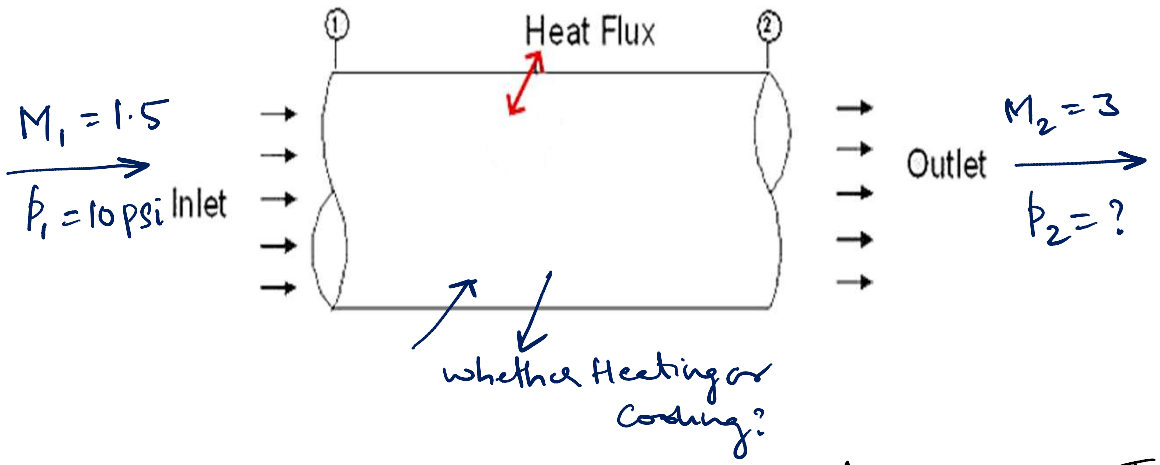
$T_2 = T^* = 1391.3$ K

$q^* = c_p (T_0^* - T_{01})$

$= 1005 (1668.5 - 290.3)$

$q^* = 1385.1$ KJ/Kg

For $M_1 = 1 \cdot 5$, $P_1 = 10$psi, $M_2 = 3$. Find P_2 and direction of heat transfer?



M_1 is supersonic and $M_2 > M_1$

$\Rightarrow$ Cooling

For $M_1 = 1.5$ Rayleigh flow tables $\rightarrow$ $\frac{P_1}{P^*} = 0.578$ $\frac{T_{01}}{T_0^*} = 0.909$

for $M_2 = 3$ Rayleigh flow tables $\rightarrow$ $\frac{P_2}{P^*} = 0.176$ $\frac{T_{02}}{T_0^*} = 0.654$

$T_{01} = 0.909\ T_0^*$

$T_{02} = 0.654\ T_0^*$

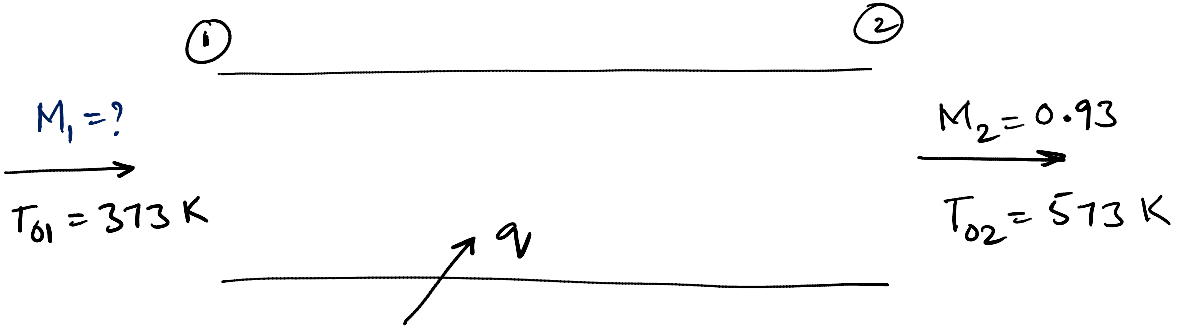
$T_{02} < T_{01}$

$\Rightarrow$ Cooling or Heat removal

- $P_2 = \frac{P_2}{P^*} \frac{P^*}{P_1} P_1 = 0.176 \times \frac{1}{0.578} \times 10 =$ psi

For $M_2 = 0 \cdot 93$, $T_{02} = 300^0C$, $T_{01} = 100^0C$. Find $M_1, \frac{P_2}{P_1} = ?$

- we know that p^*, T^*, p_0^*, T_0^* are constant in the flow



- For $M_2 = 0.93$ $\xrightarrow{\text{Rayleigh flow tables}}$ $\frac{T_{02}}{T_0^*} = 0.996$, $\frac{p_2}{p^*} = 1.085$ $\qquad \frac{p_2}{p_1} = ?$

$$\Rightarrow T_0^* = 577\ K$$

- $\frac{T_{01}}{T_0^*} = \frac{373}{577} = 0.646$
- For $\frac{T_{01}}{T_0^*} = 0.646$ $\xrightarrow{\text{Rayleigh flow tables}}$ $M_1 = 0.48$ $\qquad \frac{p_1}{p^*} = 1.815$
- $\frac{p_2}{p_1} = \frac{p_2}{p^*} \cdot \frac{p^*}{p_1} = 1.085 \times \frac{1}{1.815} = 0.6$